Python Assignment 1: Data Structures - Strings & Tuples

Strings (Concatenation, Slicing and Other methods) :

```
[4]
✓ 4s
string1="Hello "
name=input("enter the name:")

string2=string1 + name
print(string2)

string3=", welcome to Python programming"
final_string=string2+string3
print(final_string)
```

```
enter the name:Zara
Hello Zara
Hello Zara, welcome to Python programming
```

String Slicing and Indexing:

```
text= "Hello Zara, welcome to Python programming"

print("a.first character:",text[0])
print("b.last character:",text[-1])
print("first five character:",text[:5])
print("d. Last 11 characters:", text[-11:])
print("e. Reverse string:", text[::-1])

start = text.index("Python")
print("f. Word Python:", text[start:start + 6])
```

```
a.first character: H
b.last character: g
first five character: Hello
d. Last 11 characters: programming
e. Reverse string: gnimmargorp nohtyP ot emoclew ,araZ olleH
f. Word Python: Python
```

String Methods:

```
strM = "Python beginner tutorial"

print("a. Uppercase:", strM.upper())
print("b. Lowercase:", strM.lower())
print("c. Capitalize:", strM.capitalize())
print("d. Count of character t:", strM.count("t"))
print("e. Replace Python:", strM.replace("Python", "Machine Learning"))
```

```
a. Uppercase: PYTHON BEGINNER TUTORIAL
b. Lowercase: python beginner tutorial
c. Capitalize: Python beginner tutorial
d. Count of character t: 3
e. Replace Python: Machine Learning beginner tutorial
```

Tuples (Creation, Modification and Access) :

```
t1 = (10, 20, 30)
t2 = (40, 50, 60)

t_combine = t1 + t2
print("a. Combined tuple:", t_combine)


print("b. Repeated tuple:", t_combine * 3)


print("c. Third element:", t_combine[2])


print("d. First three elements:", t_combine[:3])


print("e. Last three elements:", t_combine[-3:])
```

```
a. Combined tuple: (10, 20, 30, 40, 50, 60)
b. Repeated tuple: (10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
c. Third element: 30
d. First three elements: (10, 20, 30)
e. Last three elements: (40, 50, 60)
```